

RUODI (TINAH) YUAN, BASc

Canadian Citizen | ☎ +1(778)682-7471 | @ ruodi.yuan@mail.utoronto.ca |  LinkedIn | 📍 Toronto, Canada | [Personal Portfolio](#)
Seeking internships from May 2027 to Sept 2027

EDUCATION

University of Toronto St. George

BASc in Mechanical Engineering + PEY Co-op

Toronto, Canada
Sept 2025 – Apr 2030 (expected)

University of Toronto Scarborough

Honours Bachelor of Science; Life Sciences (Health Sciences Stream); cGPA: 3.90/4.00 | University of Toronto Scarborough Dean's List

Toronto, Canada
Sept 2024 – Apr 2025

FUNCTIONAL EXPERIENCE

- Designed and validated Formula SAE drivetrain systems using SolidWorks for 3D modeling and assembly integration; supported AWD EV architecture development, PM motor assembly, rotor alignment tooling, and mechanical packaging of motor mounts, shafts, bearings, and cooling hardware.
- Developed simulation and analysis workflows using MATLAB, Simulink, and ANSYS; digitized motor efficiency maps, built lookup-table tools, evaluated endurance-event heat loads, and validated drivetrain and cooling components under representative operating conditions.
- Produced GD&T-compliant drawings, cost analysis, and manufacturing documentation for machined drivetrain components; applied DFM/DFA, tolerance planning, supplier-ready drawing practices, and fabrication constraints to support machining and assembly.
- Designed and iterated aerospace and robotic mechanisms using SolidWorks, including payload pickup/release systems, landing gear integration concepts, custom robot housings, and 3D-printed prototypes validated through hands-on testing and refinement.

SAE EXPERIENCES

University of Toronto Formula SAE Racing

Vehicle Dynamics Lead

Toronto, Canada
June 2026 - Present

- Led vehicle dynamics simulation and performance optimization using VI-CarRealTime, MATLAB, and Adams Car, analyzing lap-time sensitivity, vehicle response, and handling performance across multiple vehicle configurations.
- Conducted torsional stiffness sensitivity studies, evaluating effects on vehicle performance through lap time, lateral load transfer, steering demand, and transient vehicle dynamics metrics.
- Developed and maintained vehicle dynamics analysis workflows in MATLAB, including automated simulation post-processing, parameter sweeps, and performance visualization to support setup and design decisions.
- Performed suspension, and steering analysis, including suspension-link load correlation between Adams Car simulations and vehicle dynamics models within 20% difference to validate predicted loads.
- Supported DIL testing and validation, coordinating test parameters, and maintaining test workflows
- Maintained and compiled vehicle dynamics software PRs, supporting integration, validation, and deployment of simulation and testing updates.

Drivetrain Team Member → Drivetrain Team Lead

Sept 2024 - June 2026

- Co-led drivetrain architecture development for the team's first AWD Formula SAE EV race car and performed PM motor assembly.
- Identified rotor alignment as a key challenge in PM motor assembly due to strong magnetic attraction forces between the rotor and stator; researched industry assembly practices and designed a custom alignment tool to enable safe, precise, and repeatable rotor installation.
- Produced GD&T-compliant mechanical drawings for the inboard motor mount, planet shafts, and motor bearing, along with cost analysis and manufacturing process documentation, to support accurate fabrication and machining.
- Designed a comprehensive dual inverter cooling system using Simulink, optimizing thermal management for enhanced performance and efficiency, with SolidWorks for CAD and validated components with ANSYS FEA to ensure structural integrity under operating loads.
- Sized the system from a heat load estimate by digitizing the motor's power-efficiency map (Fischer) into a high-resolution lookup table using automated MATLAB tools and integrating it into a Simulink vehicle model to evaluate motor efficiency continuously through a simulated endurance event, establishing the thermal boundary condition for the cooling system simulation.
- Participated in Formula SAE Michigan 2026, delivering design presentations on drivetrain architecture and engineering decisions to industry judges as part of the competition's design judging event.

University of Toronto Aerospace Team

Unmanned Aerial Systems – SAE Payload Specialist

Toronto, Canada
June 2026 - Present

- Responsible to lead the design of an autonomous moving payload for the SAE Aero Design Advanced Class mission, focusing on lightweight mechanical systems and reliable autonomous operation.
- Developing navigation, capture, and release mechanisms through CAD, rapid prototyping, and engineering trade studies to maximize mission reliability.

Unmanned Aerial Systems – SAE Structure Member

Sept 2025 - June 2026

- Designed and iteratively refined payload pickup and release mechanisms for a hybrid V/STOL aircraft using SolidWorks, validating designs through 3D-printed prototypes to improve landing accuracy and reduce mechanical instability during deployment
- Collaborated cross-functionally with control and avionics teams to align mechanical design with design requirements.
- Contributed to landing gear design and fuselage structural integration, designing for accessibility to simplify maintenance and gaining hands-on experience in composite and mechanical assembly.

ADDITIONAL EXPERIENCES

UofTHacks 13 (Project Link)

2nd Place – Hack the Human-Robot Experience

Toronto, Canada
Jan 2026

- Built Red Lamp, a companion robot for students addressing isolation and burnout by providing emotionally aware encouragement during independent study sessions.
- Integrated hardware systems using a LeLamp kit, Raspberry Pi, sensors, LEDs, and custom 3D-printed components, transforming a standard robot into a responsive, companion-focused device.
- Debugged and optimized hardware–software interaction by troubleshooting power instability, loose wiring, and faulty components through voltage testing and iterative assembly, achieving reliable real-time responsiveness.
- Adapted system architecture by incorporating Arduino components after Raspberry Pi connectivity failures, improving system stability and overall performance under real-world constraints.

2025 Engineering Business Future Case Competition

3rd Place Winning Team

Toronto, Canada
Oct 2025

- Worked 3 other engineering students to develop a marketing campaign for Vick's Early Defense Nasal Spray, aiming to increase household penetration by 10% in a year.
- Identified a key gap – lack of exposure and reliability – and built strategy around tackling it using frameworks such as Porter's Five Forces and PESTLE.
- Delivered an evidence-based presentation that impressed judges with its clarity, feasibility, and analytical depth.

TECHNICAL SKILLS

- Programming & Tools:** Python, C, C++, Arduino, JavaScript, HTML, MySQL, Microsoft Office Suite.
- Engineering & Technical:** SolidWorks, MATLAB, Simulink, Catia, ANSYS, machining and manufacturing processes, Vi-Grade CarRealTime, rFpro, ADAMS Car.
- Language:** English(native), Chinese(native), Japanese(elementary).

LICENSES & CERTIFICATIONS

Certified 3DEXPERIENCE Mechanical & Shape Designer - CATIA Associate – Issued by Dassault Systèmes

Certified SOLIDWORKS Associate (CSWA) – Issued by Dassault Systèmes

Basic Machining: Lathe, Mill, Drill Press – Issued by George Brown College

July 2026

Dec 2025

Oct 2025